DOAN (JACKSON) NGUYEN

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EDUCATION

University of California, Irvine, Henry Samueli School of Engineering, Irvine, CA — June 2027
B.S., Electrical Engineering | GPA: 3.594 | Specialization in Electronic Circuit Design

Santa Ana College, Santa Ana, CA — August 2023 - June 2025
A.S., Engineering, High Honors | GPA: 3.960

ENGINEERING PROJECT EXPERIENCE

Zotbotics Level 2 Project: 3D Robot Arm — March 2026 - Present
Electrical Systems & PCB Designer

- Design circuit schematics and 2-layer PCB layouts in KiCad, selecting components based on expected electrical performance, actuator requirements, and system constraints.
- Simulate circuit behavior in LTspice using selected component values to verify expected voltage, current, and signal behavior before hardware assembly and PCB layout.
- Design and assemble breadboard wiring prototypes from technical wiring diagrams to support firmware testing before mechanical integration.
- Characterize, validate, and troubleshoot a 24V/15A power distribution network for the system; utilize multimeter diagnostics to verify stable voltage across critical nodes and reduce risk of electrical failure.
- Research and implement control logic for various actuators and sensors, including servo motors, NEMA stepper motors, and IR remote interfaces on Arduino Mega 2560 using Arduino IDE.
- Collaborate with mechanical teammates to integrate electrical and control systems into a 3D-printed PLA robotic arm.

Irvine Hacker Fab (Spin Coater Team) — January 2026 – June 2026
PCB Designer

- Designed custom 2-layer PCBs in KiCad to replace point-to-point wiring with consolidated board-to-board interfaces; ensuring stable connections between ESP32 and peripherals and increasing system reliability.
- Collaborated with embedded systems teammates to translate physical pin maps into formal schematics and wiring diagrams to support reliable hardware integration.
- Assembled and soldered components on prototype boards to maintain stable signal and power transmission across 20+ microcontroller and peripheral pins.

UCI OAI Summer Bridge: Mobile Robotic Car Project — August 2025
Electrical Team Member

- Wired, soldered, and assembled electrical control systems for the mobile platform while coordinating with mechanical team members to integrate electronic controls into the device housing.
- Developed and debugged Arduino-based navigation logic using ultrasonic sensors and DC motor control algorithms in Arduino IDE, resulting in precise obstacle avoidance and path correction.
- Tested and refined hardware and firmware interactions to ensure reliable performance of the robotic control system.

WORK EXPERIENCE

Santa Ana College Math Center, Santa Ana, CA — September 2024 – December 2024
Math Tutor

- Tutored 30+ students per week in algebra and calculus through one-on-one and group sessions.
- Explained technical problem-solving methods clearly to improve student understanding and confidence.

ACTIVITIES

Science Olympiad 2025-2026 — October 2025 – February 2026
Volunteer at OC Regional tournament

- Led 2+ volunteers in managing the test room, communicating with participants in written and device tests, grading and managing data entry for the exam results of 40+ participants following strict deadlines of the competition.
- Assisted in distributing and labeling test materials and devices from inventory to packets of 50+ events.
- Co-developed written test materials, including test packets, answer sheets, and answer keys.

SKILLS

Electrical Design: KiCad, LTspice, Cirkit Designer IDE, circuit analysis, 2-layer PCB, power distribution, schematic capture
Lab Equipment: Multimeter diagnostics, oscilloscope basics, function generator basics, power supplies
Embedded Systems: Arduino, servo motors, stepper motors, DC motor control, ultrasonic sensors
Fabrication: Soldering, Wiring, Breadboarding, Crimping, 3D Printing, Electrical Assembly
Software & Programming: Arduino IDE, MATLAB, GitHub, Microsoft Office, C/C++